

Q1.

Solve the differential equation

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 10y = 26e^x$$

given that $y = 5$ and $\frac{dy}{dx} = 11$ when $x = 0$. Give your answer in the form $y = f(x)$.

(Total 10 marks)

Q2.

Find the general solution of the differential equation

$$\frac{d^2y}{dx^2} + 4y = 8x^2 + 9 \sin x$$

(Total 8 marks)

Q3.

- (a) Find the value of the constant k for which $k \sin 2x$ is a particular integral of the differential equation

$$\frac{d^2y}{dx^2} + y = \sin 2x$$

(3)

- (b) Hence find the general solution of this differential equation.

(4)

(Total 7 marks)

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Q4.

It is given that y satisfies the differential equation

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 5y = 8 \sin x + 4 \cos x$$

- (a) Find the value of the constant k for which $y = k \sin x$ is a particular integral of the given differential equation.

(3)

- (b) Solve the differential equation, expressing y in terms of x , given that $y = 1$ and $\frac{dy}{dx} = 4$ when $x = 0$.

(8)

(Total 11 marks)

Q5.

- (a) Find the general solution of the differential equation

$$\frac{d^2 y}{dx^2} + 4 \frac{dy}{dx} + 5y = 5$$

(6)

- (b) Hence express y in terms of x , given that $y = 2$ and $\frac{dy}{dx} = 3$ when $x = 0$.

(4)

(Total 10 marks)

Q6.

- (a) Find the general solution of the differential equation

$$\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} - 3y = 10e^{-2x} - 9$$

(10)

- (b) Hence express y in terms of x , given that $y = 7$ when $x = 0$ and that $\frac{dy}{dx} \rightarrow 0$ as $x \rightarrow \infty$.

(4)

(Total 14 marks)

Q7.

- Find the general solution of the differential equation

$$\frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} + 3y = 6 + 5 \sin x$$

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(Total 12 marks)

Q8.

- (a) Find the roots of the equation $m^2 + 2m + 2 = 0$ in the form $a + ib$.

(2)

- (b) (i) Find the general solution of the differential equation

$$\frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + 2y = 4x$$

(6)

- (ii) Hence express y in terms of x , given that $y = 1$ and $\frac{dy}{dx} = 2$ when $x = 0$.

(4)

(Total 12 marks)

Q9.

It is given that y satisfies the differential equation

$$\frac{d^2 y}{dx^2} - 5 \frac{dy}{dx} + 4y = 8x - 10 \cos 2x$$

- (a) Show that $y = 2x + \sin 2x$ is a particular integral of the given differential equation.

(3)

- (b) Find the general solution of the differential equation.

(4)

- (c) Hence express y in terms of x , given that $y = 2$ and $\frac{dy}{dx} = 0$ when $x = 0$.

(4)

(Total 11 marks)

Q10.

The motion of a particle, which moves along a straight line, is governed by the differential equation

$$2 \frac{d^2 x}{dt^2} + 6 \frac{dx}{dt} + 9x = 0$$

where x is the displacement of the particle from the origin at time t .

Initially the particle is at rest at the point where $x = a$.

- (a) Find x in terms of a and t .

(9)

- (b) Is the damping of the motion of the particle light, critical or heavy? Give a reason for your answer.

(2)

(Total 11 marks)